

VAA interiors + architecture
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April 6, 2011

Ms. Dana Truscott
Director of Corporate Real Estate
Ardsley Financial Group
Corbin Tower, Level 19
Los Angeles, California

By email and courier

Re: Project 632588 — Ardsley Financial Group Headquarters, Corbin Tower, Levels 12–19
Substitution Request SR-014 — proposed deletion of Specification Section 09 05 61, Moisture Vapor Mitigation — VAA's recommendation

Dear Ms. Truscott,

This letter states VAA's formal position on SR-014, as you requested at Project Meeting No. 47.

VAA recommends against deleting Section 09 05 61.

1. The measured condition of the slab.

Callenbach Materials Testing installed fourteen in-situ relative humidity probes on Levels 12 through 19 in accordance with ASTM F2170 and reported results on February 8, 2011 (Report CMT-11-0412). Readings ranged from 92.1 to 96.4 percent relative humidity. Twelve of the fourteen probes read above 93 percent. The specification threshold for installing adhered floor coverings is 85 percent. No test area on the eight floors met it, and the lowest reading on the Project was seven points above the threshold.

2. Why we do not accept the calcium chloride results as an answer.

We have read Prewitt Building Science's memorandum of March 9, 2011. We do not dispute Prewitt's numbers. We dispute what they can be used to conclude.

ASTM F1869 measures vapor emission from approximately the upper half inch of the slab. ASTM F2170 measures relative humidity within the body of the slab at 40 percent of its depth. On a lightweight concrete fill over composite metal deck, which can only dry through its upper face, the surface skin dries first and dries fastest. A low calcium chloride result on this substrate is the expected result and it is not evidence that the slab is dry. It is evidence that the top of the slab is dry.

The moisture that will reach the adhesive over the service life of this installation is the moisture presently sitting at depth, and that moisture is at 92 to 96 percent relative humidity. Section 09 05 61 Article 1.2.D excludes F1869 on this Project for precisely this reason, and it was written that way before either test was performed.

We would add one point that Prewitt's memorandum does not address. The 5.0 pound limit Prewitt measures against is the limit published for the carpet tile pressure sensitive adhesive.

The resilient tile and the heat-welded resilient sheet scheduled for this Project carry a published limit of 3.0 pounds. Five of Prewitt's own nine kits returned 3.0 or above. On the products actually scheduled in those areas, Prewitt's own data does not support Prewitt's own conclusion.

3. What we expect will happen if the system is deleted.

Adhered resilient flooring and carpet installed over a slab in this condition, with no vapor barrier, will in our professional judgment fail. The mechanism is alkaline hydrolysis: moisture migrating upward through the slab carries alkali to the adhesive layer, the adhesive saponifies, bond is lost, and the flooring lifts at seams and edges. Standing moisture beneath a failed assembly creates a risk of microbial growth and associated indoor air quality complaints.

We want to be precise about the timing, because it is the most important thing in this letter. **This failure is not immediate.** The installation will look correct at turnover, it will look correct at the end of the warranty period, and it will most likely look correct for a decade. In our experience with this substrate and this condition, visible failure across the general office areas appears **between eight and fifteen years** after installation. By then every warranty available on this Project will have expired and the remediation will be an unbudgeted capital event carried by whoever owns the building.

Two qualifications to that window, because they matter. **Heat-welded resilient sheet in wet-use rooms fails much sooner**, typically within the first two to four years, because the covering is effectively impermeable and the room adds its own water at the surface. On this Project that means the pantries and the Level 19 wellness room. And any area kept under continuously closed furniture or equipment will fail earlier than the open floor around it.

4. The economics as we see them.

The credit is real and we are not disputing it. SR-014 returns \$347,900 and 18 calendar days.

Our view is that \$347,900 today is being traded against a remediation that will cost a multiple of it later, because remediation means removing and replacing the flooring across occupied floors, mitigating the slab at that time anyway, and relocating staff while it happens. We would not make that trade. We recognize that we are not the party making it.

5. Where this leaves us.

The decision is the Owner's. Section 09 05 61 is a betterment protecting the Owner's asset, and an owner is entitled to decline a betterment after being advised.

If Ardsley elects to proceed with the deletion, VAA will issue the necessary Architect's Supplemental Instruction. **We will issue it only upon written direction from Ardsley that records the deletion as owner-directed and made over VAA's written recommendation.** We ask for that in writing not to be difficult but because a decision of this kind should be documented by the party making it, and because the people who will encounter this assembly are not likely to be any of us.

Please confirm Ardsley's direction in writing.

Respectfully,

Roland Ashcombe

Roland Ashcombe, AIA, LEED AP
Principal in Charge

VAA interiors + architecture

cc: Bernard Kilcoyne, Ardsley Financial Group
Owen Radcliffe, Thackston Bell Construction
Marguerite Okonkwo, AIA, VAA interiors + architecture
Project file 632588